

Hermes F&O Trading Master Guide

A practical curriculum for Futures & Options trading, quantitative algorithms, predictive modelling, risk management, and training Hermes for disciplined automated trading.

Important: This guide is educational. No algorithm can reliably predict prices with certainty, and leveraged F&O trading can cause rapid and substantial losses. The objective is to build a tested statistical edge with strict risk controls—not a guaranteed-profit machine.

1. What Successful F&O Trading Means

- Define success using expectancy, not prediction accuracy. A strategy is viable only when expected gross edge remains positive after brokerage, exchange charges, taxes, spread, slippage and market impact.
- Expectancy: $E[P\&L] = P(\text{win}) \times \text{AvgWin} - P(\text{loss}) \times \text{AvgLoss} - \text{Costs}$.
- Key metrics: expectancy, profit factor, maximum drawdown, Sharpe/Sortino, tail loss, MAE (maximum adverse excursion) and MFE (maximum favorable excursion).
- Do not build an expiry-day lottery machine. Build a risk-adjusted decision system.

2. Learn F&O Mechanics First

- Master spot, futures, calls, puts, ITM/ATM/OTM, strike, expiry, premium, intrinsic/extrinsic value, lot size, margin, leverage, settlement, rollover and futures basis.
- Learn the Greeks: Delta, Gamma, Theta, Vega and Rho.
- Understand that an option premium depends on the underlying, strike, time, interest rate, volatility and market supply/demand.

3. The Core Option Models

- Black-Scholes: $C = S \cdot N(d1) - K \cdot e^{(-rt)} \cdot N(d2)$, with $d1 = [\ln(S/K) + (r + \sigma^2/2)t] / (\sigma\sqrt{t})$, $d2 = d1 - \sigma\sqrt{t}$.
- Limitation: real markets exhibit volatility smiles/skews and changing volatility, so Black-Scholes should be treated as a baseline rather than an oracle.
- Heston: models stochastic variance and can be calibrated to a volatility surface.
- SABR: useful for modelling volatility smiles and fast surface fitting.
- Monte Carlo: simulate distributions of future paths and estimate target/stop probabilities, expected $P\&L$ and tail risk.

4. Classical Trading Algorithms

- Trend / moving averages: use EMA/SMA alignment as a regime filter rather than a standalone entry.
- Breakout: enter only when price breaks a defined range and confirmation includes volume, ATR, VWAP and/or futures confirmation.
- Pullback: identify an established trend, wait for a controlled retracement, then enter on structural reclaim.
- Mean reversion: use standardized distance such as $Z = (P - \mu) / \sigma$, but only when the market is demonstrably range-bound.

- Time-series momentum: $\text{signal} = \text{sign}(P_t - P_{t-N})$; combine several horizons rather than relying on one lookback.
- VWAP: $\text{VWAP} = \Sigma(\text{Price} \times \text{Volume}) / \Sigma \text{Volume}$. Track reclaim, rejection, slope and deviation.
- ATR: use volatility-scaled stops and position sizing instead of fixed point distances.

5. Market-Regime Detection

- Use a regime engine before selecting a strategy. Useful states include low-volatility trend, high-volatility trend, low-volatility range, high-volatility range, panic and transition.
- Hidden Markov Model (HMM): infer hidden market states from observed price/volatility/flow features.
- GARCH: estimate conditional volatility and volatility clustering. It is especially useful for expected movement and position sizing.
- Strategy selection should be conditional: trend → momentum/breakout; range → mean reversion; high IV → volatility structures; panic → reduce risk or do nothing.

6. Options Intelligence: IV, OI, Skew and Basis

- IV vs RV: compute $\text{VRP} = \text{IV} - \text{realized volatility}$. Compare implied movement with expected realized movement rather than simply calling IV 'high' or 'low'.
- Volatility surface: model $\text{IV}(K, T)$ across strikes and expiries. Monitor skew, smile, term structure, IV spikes and relative inconsistencies.
- Put-call ratio: $\text{PCR} = \text{Put OI} / \text{Call OI}$. Never use a simplistic 'PCR high = buy' rule; combine OI changes with price, IV and volume.
- OI interpretation: price↑ + OI↑ can indicate long buildup; price↓ + OI↑ can indicate short buildup; price↑ + OI↓ can indicate short covering; price↓ + OI↓ can indicate long unwinding. Treat these as inferences, not certainty.
- Futures basis: $\text{Basis} = \text{Futures} - \text{Spot}$. Track normalized basis, historical extremes and convergence into expiry.

7. Advanced 'Cheat-Like' Predictive Information

- Order-book imbalance: $\text{OBI} = (\text{BidVolume} - \text{AskVolume}) / (\text{BidVolume} + \text{AskVolume})$. Static book size is insufficient; Hermes should learn how queues appear, disappear, replenish and get consumed.
- Order-flow imbalance (OFI): combine aggressive buys/sells, new orders, cancellations and queue changes.
- Microprice: $(\text{Ask} \times \text{BidSize} + \text{Bid} \times \text{AskSize}) / (\text{BidSize} + \text{AskSize})$. It can be a useful short-horizon reference beyond the simple midpoint.
- DeepLOB-style modelling: use convolutional layers for order-book spatial structure and recurrent/attention layers for temporal structure. Predict probabilities of up/down/flat over a short horizon rather than an exact future price.
- Dealer/gamma exposure: estimate aggregate gamma exposure to identify environments where hedging flows may amplify or dampen moves. Use as a regime/context signal, never as a guarantee.
- Lead-lag: test whether GIFT NIFTY, global futures, USDINR, crude, sector indices, breadth or heavyweight stocks consistently lead the target instrument. Correlation alone is not enough.

8. Statistical Arbitrage

- Pairs / spread: $\text{Spread} = PA - \beta PB$. Test stationarity/cointegration before trading.
- Z-score: $Z = (\text{Spread} - \mu) / \sigma$. Extreme values can provide mean-reversion candidates if the relationship remains stable.
- Kalman filter: estimate time-varying hedge ratios β_t rather than assuming a permanent relationship.

9. Machine Learning Stack for Hermes

- Tier 1: Logistic Regression, Ridge/Lasso, Random Forest, Gradient Boosting, XGBoost/LightGBM.
- Tier 2: LSTM, GRU, Temporal CNN and Transformers.
- Specialized: DeepLOB-style models, temporal fusion architectures and, later, reinforcement learning.
- Do not train Hermes to output 'NIFTY will be 24,721.53'. Prefer probability targets such as $P(\text{Return5m} > \text{threshold})$, multi-class up/flat/down, expected move, stop-before-target probability and realized volatility.
- Build an ensemble: price structure + order flow + volatility + IV surface + gamma + breadth + regime. A meta-model can estimate $P(\text{trade success} \mid \text{features})$.

10. Risk and Position Sizing

- Basic sizing: $\text{PositionSize} = \text{MaxRiskPerTrade} / (\text{StopDistance} \times \text{ValuePerPoint})$.
- Use fractional Kelly only if probabilities are robust; for early systems, fixed-fractional risk is easier to control.
- Stops can be structural, volatility-based (e.g. $k \times \text{ATR}$), time-based, probability-based or information-based.
- Never use martingale, uncontrolled averaging down, or doubling after losses.
- Hermes should have hard limits independent of ML: maximum daily loss, trade loss, position size, exposure, gamma/vega exposure, slippage and drawdown.

11. Option Strategy Selection

- First classify direction, volatility, expected movement and holding period. Only then select the option structure.
- Bullish + low IV $\rightarrow$ potentially call debit spread. Bullish + high IV $\rightarrow$ potentially defined-risk bullish spread. Bearish equivalents use put structures.
- Neutral + high IV may support defined-risk premium-selling structures; large expected move + relatively low IV may support defined-risk long-volatility structures.
- Choose strikes using Greeks, expected distribution, liquidity and transaction costs—not simply by picking the nearest OTM strike.

12. Execution Algorithms

- TWAP: distribute execution over time.
- VWAP: align execution with market volume.
- POV: participate at a chosen fraction of market volume.
- Iceberg: expose only part of intended size where supported.
- Almgren-Chriss: balance market-impact cost against risk from executing too slowly.

- For a retail Hermes system, liquidity-aware limit orders plus VWAP/POV-style execution are more practical than attempting HFT market making.

13. Training Hermes: Data → Features → Labels

- Data: spot, futures, options chain, OI, volume, IV, bid/ask, L2 book, trades, cancellations, India VIX, global markets, USDINR, crude, rates and event calendars.
- Features: price structure, momentum, VWAP, ATR, basis, OI dynamics, IV/skew/term structure, Greeks, gamma estimate, breadth, order-flow imbalance, microprice and cross-market lead-lag.
- Labels: 5m/15m forward return, movement magnitude, stop-before-target, realized volatility and simulated trade outcome.
- Multi-task learning: predicting direction, magnitude, volatility and risk together can produce a more useful decision layer.

14. Backtesting Without Fooling Yourself

- Use walk-forward validation. Example: train 2019–2022 → validate 2023 → test 2024; roll forward and repeat.
- Never randomly shuffle financial time series.
- Prevent look-ahead bias: every feature must use only information that was actually available at the trade timestamp.
- Model real brokerage, taxes, exchange fees, spread, slippage and market impact.
- Keep a completely untouched final test period. Do not repeatedly tune against it.

15. Hermes Deployment Pipeline

- Mode 1 — Research: no real orders.
- Mode 2 — Historical backtest: replay data with realistic costs.
- Mode 3 — Paper/live shadow: live signals, zero execution.
- Mode 4 — Small live: tiny risk after sufficient evidence.
- Mode 5 — Scale: increase exposure only after stable out-of-sample performance.
- Give Hermes a 'NO TRADE' action. A professional system is selective and should trade only when expected value exceeds costs and risk.

16. Recommended Hermes V1

- Underlying: NIFTY 50.
- Instruments: NIFTY futures plus defined-risk option spreads.
- Timeframes: 5m, 15m and 1h.
- Features: EMA structure, ATR, VWAP, volume, futures basis/OI, option OI, IV, skew, India VIX, gamma estimate, breadth, global futures and order-flow features.
- Regime: HMM or comparable probabilistic regime classifier.
- Main model: LightGBM/XGBoost probability model.
- Microstructure model: DeepLOB-style model once high-quality L2/trade data are available.

- Risk: start around 0.25–0.50% account risk per trade as a research starting point, then validate; never assume this is universally appropriate.
- Execution: liquidity-aware limit orders.
- Safety: independent risk daemon and kill switch.

17. The Hermes Architecture

- Market data → Feature engine → Regime engine → Signal models → Options/volatility models → Meta-model → Risk engine → Execution engine → Broker API.
- Signal models should include trend, breakout, mean reversion, statistical arbitrage, ML and eventually order-book models.
- The risk engine must be able to veto every model signal.

18. The 12-Month Training Curriculum

- Months 1–2: F&O mechanics, Python, probability and statistics.
- Months 3–4: price structure, technical features and honest backtesting.
- Months 5–6: futures, OI, Greeks, IV surface and options mechanics.
- Months 7–8: GARCH, Heston/SABR, Monte Carlo and volatility strategies.
- Months 9–10: order flow, L2, DeepLOB-style modelling and ML.
- Month 11: portfolio/risk/execution engine.
- Month 12: paper/live-shadow deployment and evaluation.

19. The Real 'Cheat Codes'

- 1. Predict volatility before trying to predict exact direction.
- 2. Study how the order book changes, not just its displayed size.
- 3. Monitor potential option-hedging pressure and gamma regime.
- 4. Compare implied volatility with expected realized volatility.
- 5. Detect market regime and switch strategy accordingly.
- 6. Predict a probability distribution instead of one exact price.
- 7. Make Hermes capable of refusing to trade.

20. Final Blueprint

- Information → Features → Regime → Probability → Expected Value → Risk → Execution.
- The strongest system is unlikely to be the one with the highest raw prediction accuracy. It is the one that recognizes when its information is unreliable and protects capital.
- Use academic findings as hypotheses to test, not promises of profit. Every proposed edge must survive out-of-sample testing, costs, slippage and regime changes.

Selected verification sources

SEBI — Study: Profitability of Individual Traders in the Equity Derivatives Segment (FY25–FY26), published August 20, 2026.

NSE — Equity Derivatives Contract Specifications; NSE Option Chain; Equity Derivatives Trading System; Retail Algo Trading framework and FAQs.

DeepLOB — Zhang, Zohren and Roberts, 'DeepLOB: Deep Convolutional Neural Networks for Limit Order Books', arXiv:1808.03668.

Trend/time-series momentum research and volatility-risk-premium research are included as research foundations; empirical profitability is implementation- and cost-dependent.